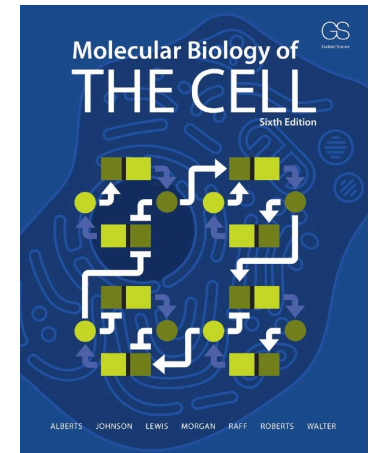


*Materials
Taken from*

Key concepts of Bioenergetics Part 2



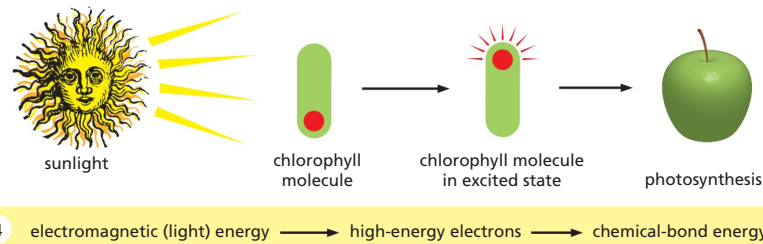
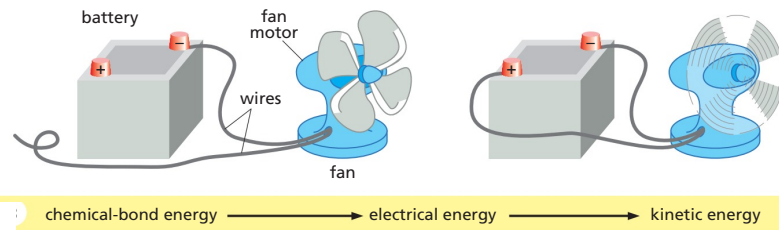
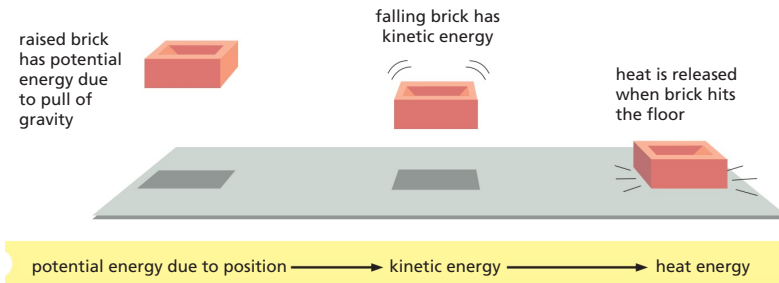
**Dr. P. Satpati,
BSBE, IIT Guwahati**

Laws of thermodynamics

1st Law : Energy is conserved, but energy can change its form

2nd law : Any spontaneous process increases the disorder (entropy) of the universe

1st Law



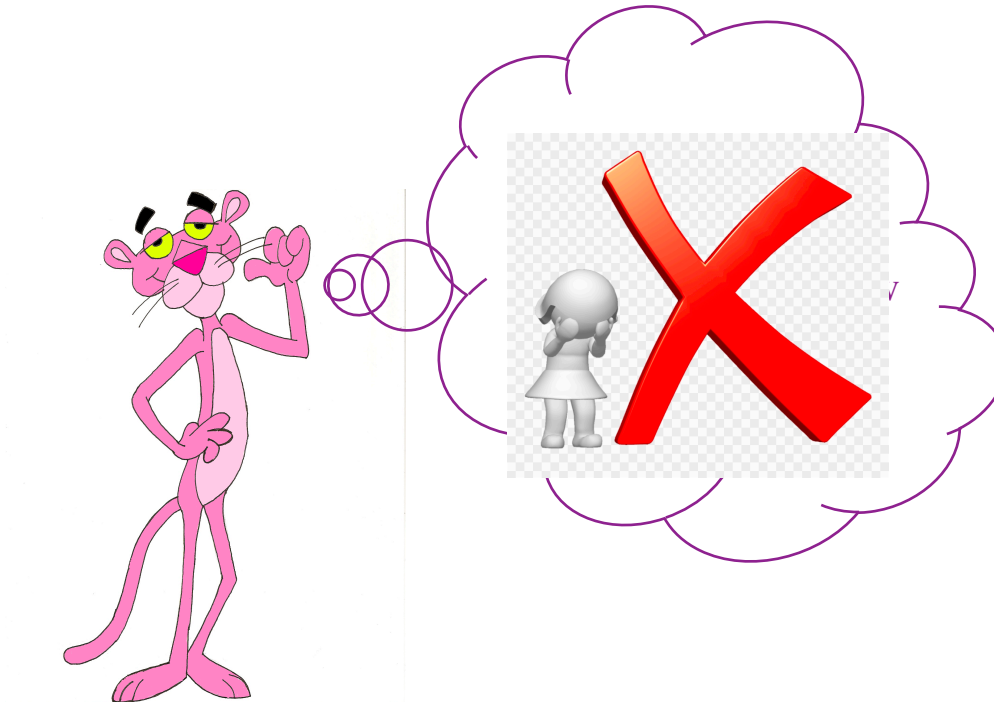
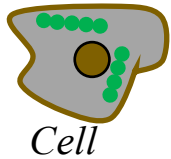
Life obey 1st law

2nd law of thermodynamics

$$\Delta S_{\text{universe}} = \Delta S_{\text{system}} + \Delta S_{\text{surroundings}} \geq 0$$

(S = entropy,
Measure of disorder)

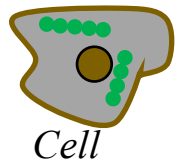
Living things: Create and maintain order.



2nd law of thermodynamics

$$\Delta S_{\text{universe}} = \Delta S_{\text{system}} + \Delta S_{\text{surroundings}} \geq 0$$

Living things: Create and maintain order.



- living organism must perform a never-ending stream of chemical reactions.
- Cell can be viewed as a tiny “*Chemical Factory*”

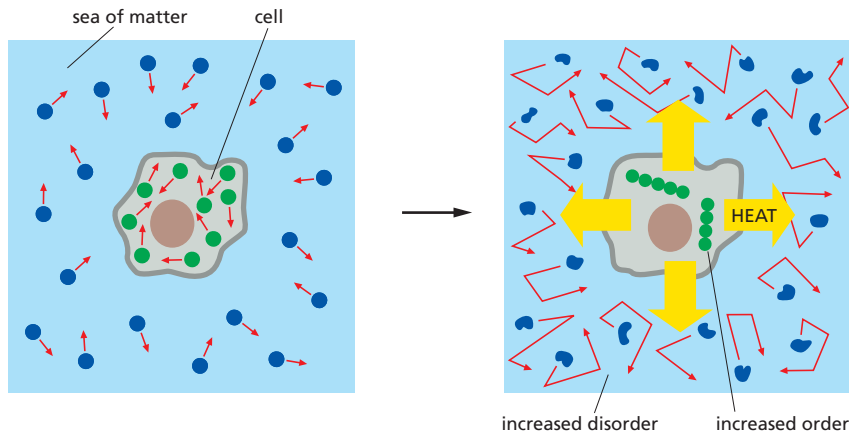
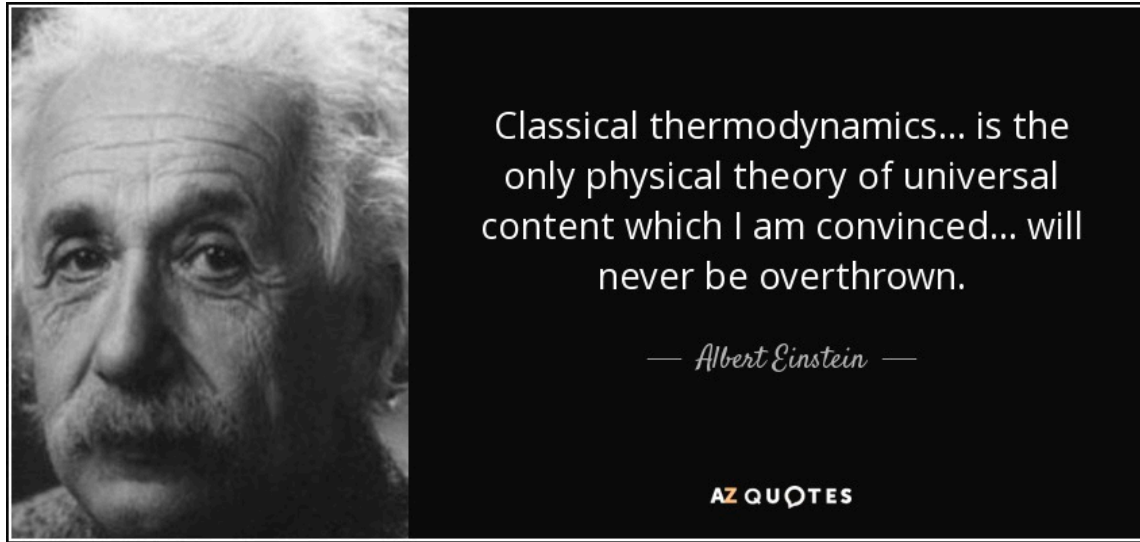


Figure 2-16 A simple thermodynamic analysis of a living cell. In the diagram on the left, the molecules of both the cell and the rest of the universe (the sea of matter) are depicted in a relatively disordered state. In the diagram on the right, the cell has taken in energy from food molecules and released heat through reactions that order the molecules the cell contains. The heat released increases the disorder in the environment around the cell (depicted by jagged arrows and distorted molecules, indicating increased molecular motions caused by heat). As a result, the second law of thermodynamics—which states that the amount of disorder in the universe must always increase—is satisfied as the cell grows and divides. For a detailed discussion, see Panel 2-7 (pp. 102–103).

**CAUTION : Never forget
”surrounding” if you want
to use the equation**

Biological order is made possible due to “lost heat” from Cells

“Lost heat” => your body temperature (37°C)



**But : Use correctly
remember "surrounding"**



Great: I do not have to study another law.

Life obey laws of thermodynamics

1st Law (energy conversion)

2nd law (create and maintain order)

2nd law

$$\Delta S_{universe} = \Delta S_{system} + \Delta S_{surroundings} \geq 0$$

(S = entropy,
Measure of disorder)

III Same statement as above but I can think only about the system (below).

$$\Delta G_{system} \leq 0 \quad (G = \text{Free energy})$$

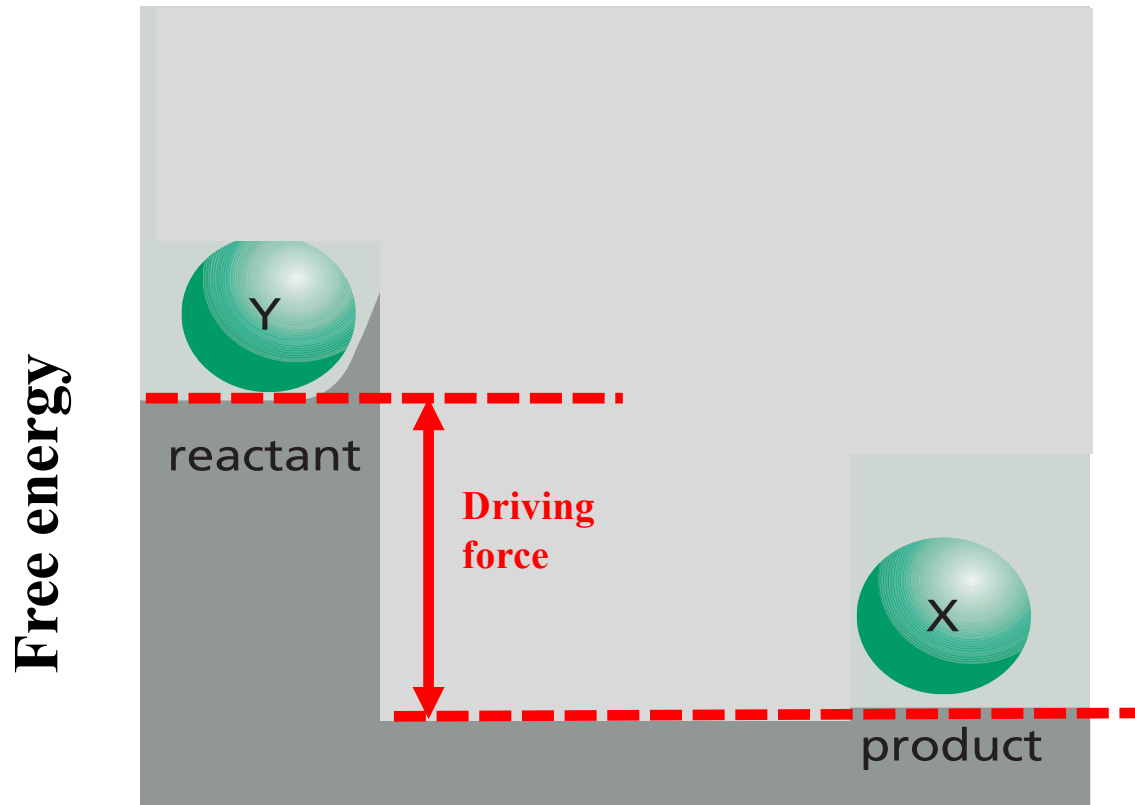
2nd law (in words)

➤ Spontaneous process increases the entropy of the universe

III

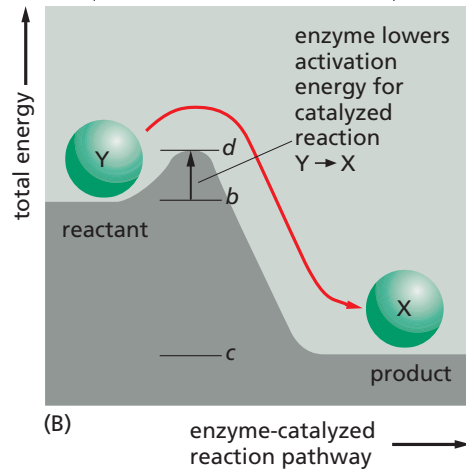
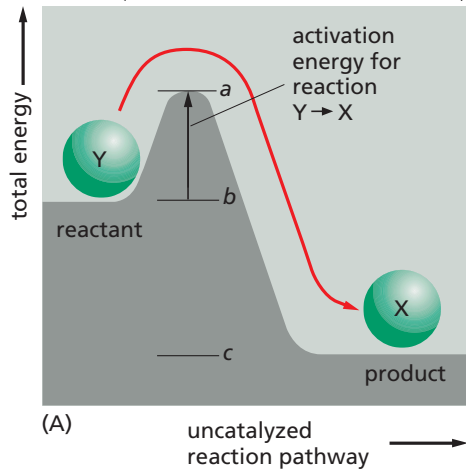
➤ Spontaneous process is the direction that goes “downhill” in free energy

Driving force = Spontaneous process

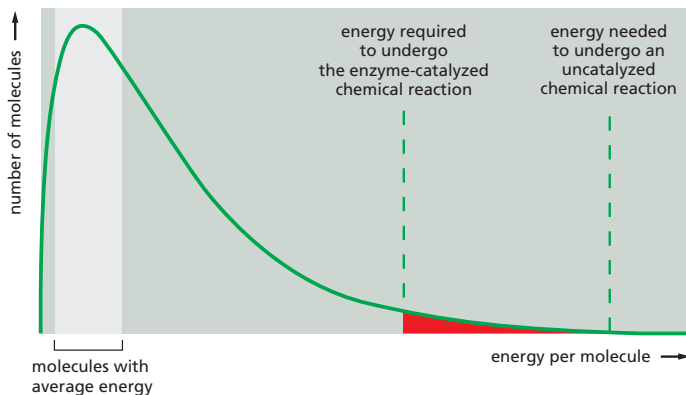


Paper + O₂
(**REACTANT**)

→ **smoke + ashes + heat + CO₂ + H₂O**
(**PRODUCT**)

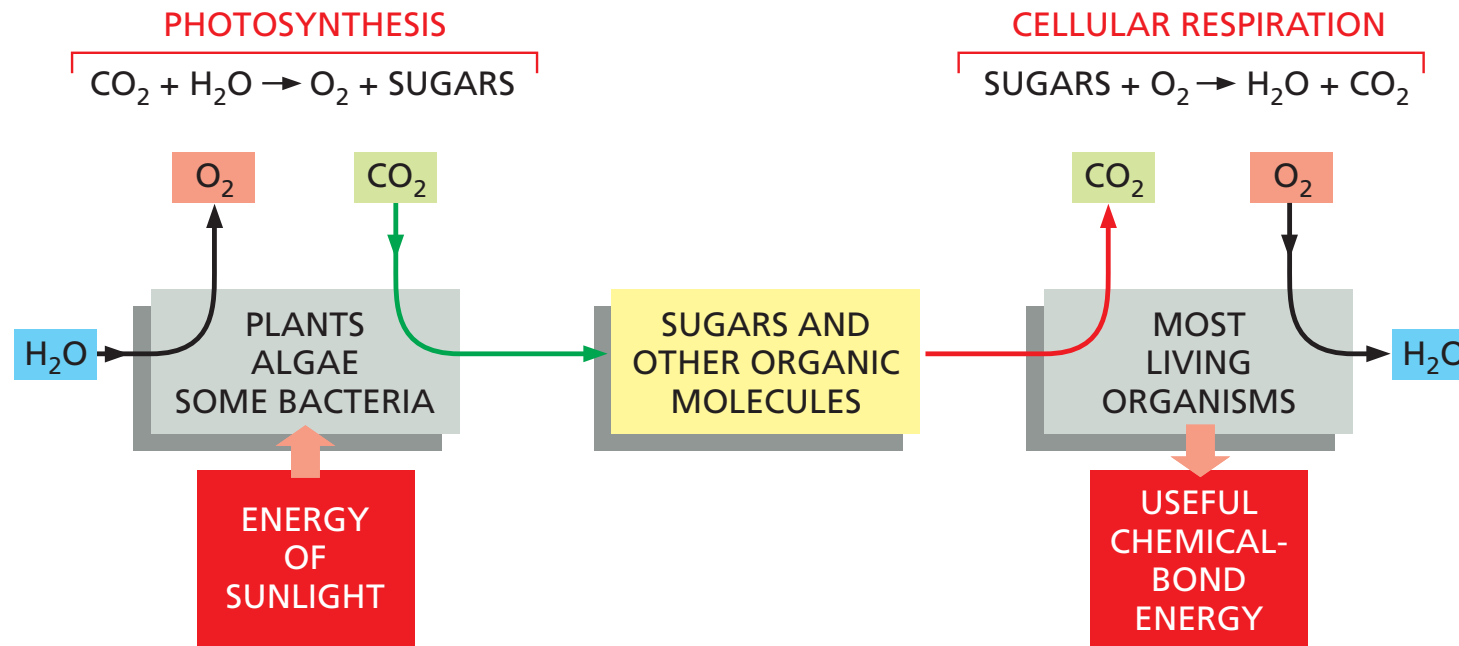


- “X” is energetically favourable than “Y”
- But “Y” require kick to climb the hill-top
- Enzyme reduces the height of the Hill-top

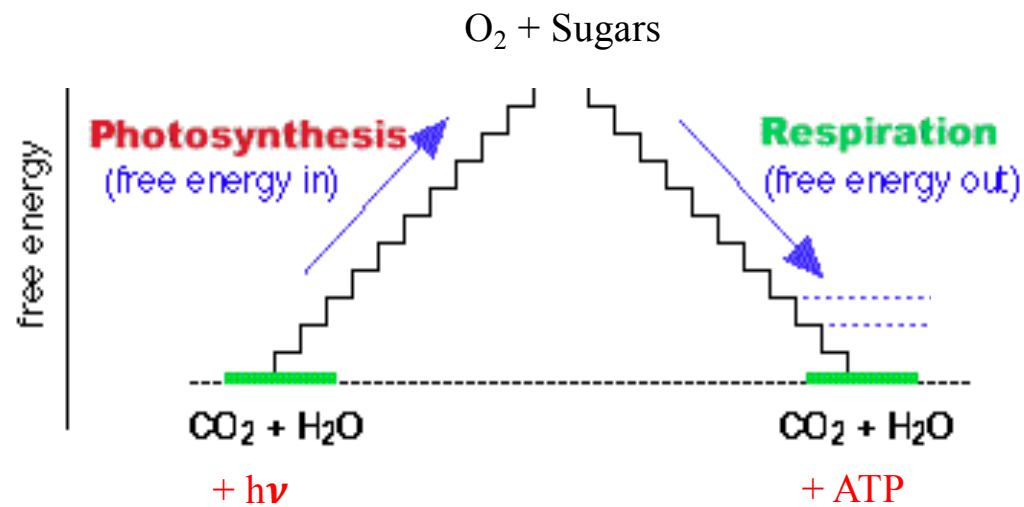
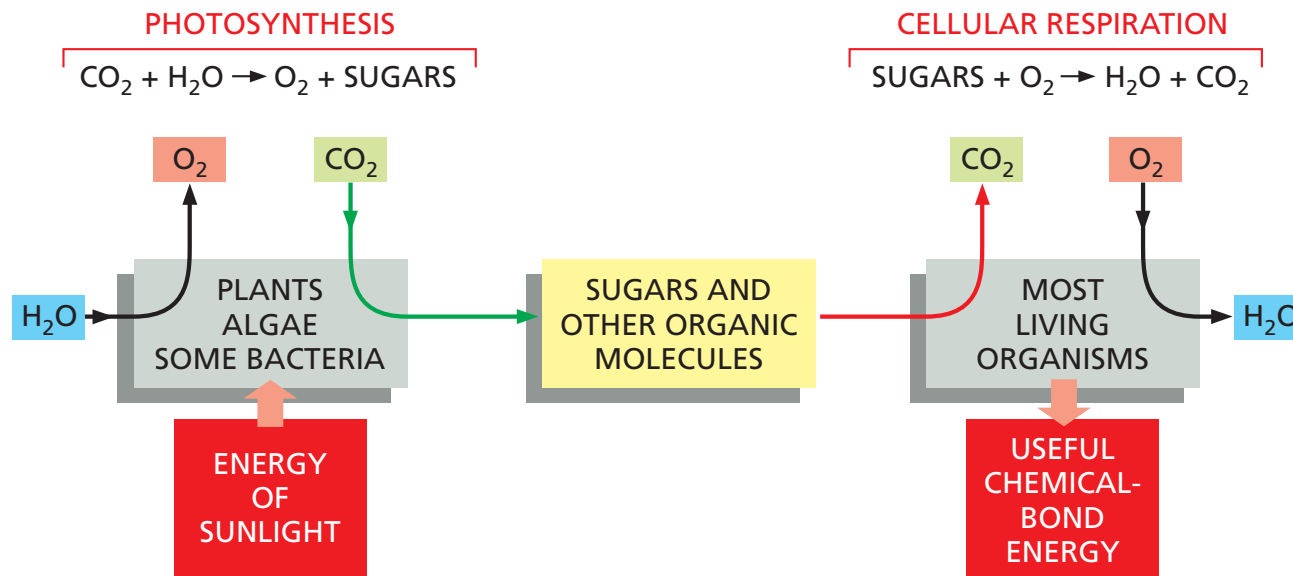


- Population of identical substrate molecules will have a range of energies, distributed as shown on the graph.
- Lowering the activation energy greatly increases the probability of a reaction.

How cell obtain energy ?



Cells obtain energy by the oxidation of organic molecules



Each cell can be viewed as a tiny chemical factory, performing many millions of reactions every second.

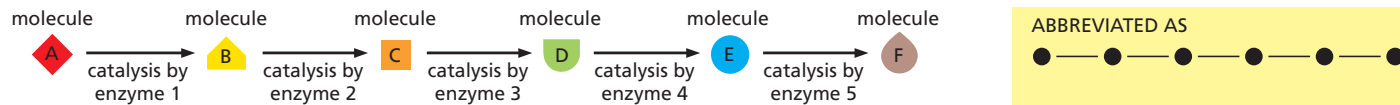


Figure 2-13 How a set of enzyme-catalyzed reactions generates a metabolic pathway. Each enzyme catalyzes a particular chemical reaction, leaving the enzyme unchanged. In this example, a set of enzymes acting in series converts molecule A to molecule F, forming a metabolic pathway. (For a diagram of many of the reactions in a human cell, abbreviated as shown, see

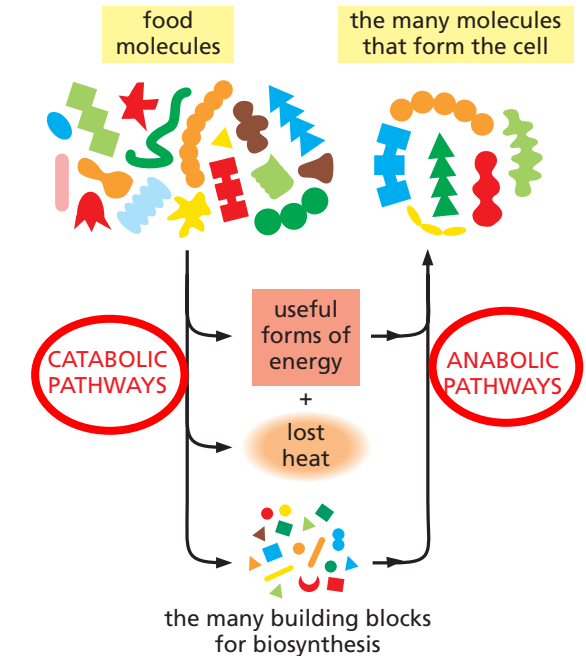


Figure 2-14 Schematic representation of the relationship between catabolic and anabolic pathways in metabolism. As suggested in this diagram, a major portion of the energy stored in the chemical bonds of food molecules is dissipated as heat. In addition, the mass of food required by any organism that derives all of its energy from catabolism is much greater than the mass of the molecules that it can produce by

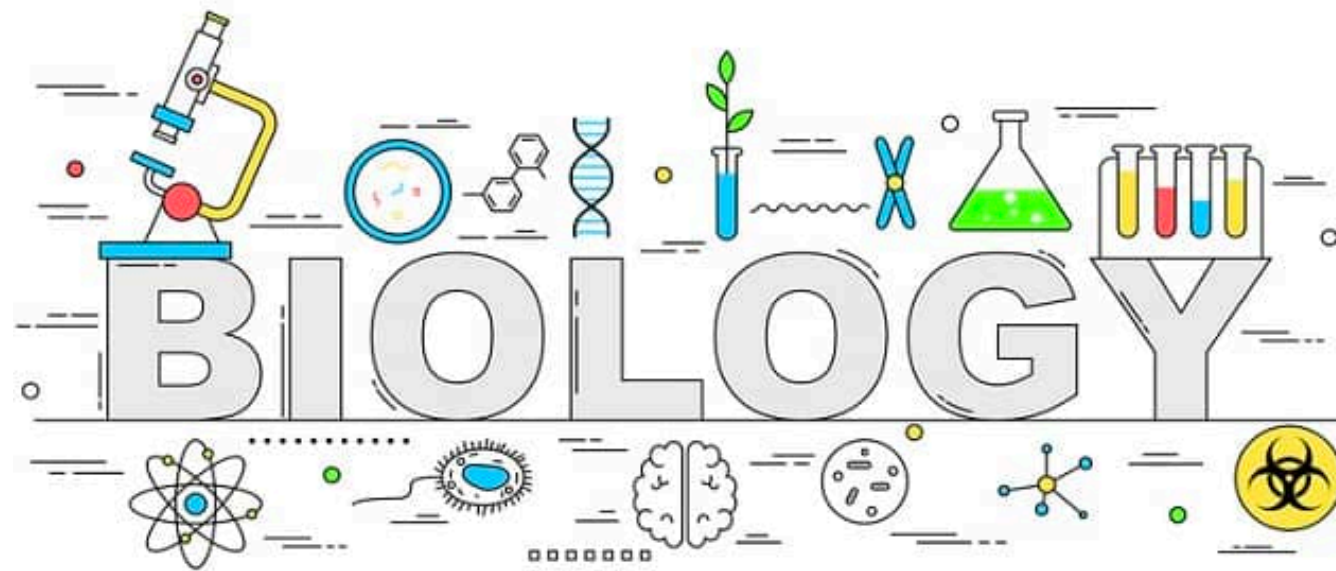
Biological catalysts = enzymes

Catabolic process = Energy producing process

Anabolic process = Energy consuming process

Take Home

1. Life obey laws of thermodynamics –
 - Energy is conserved but changes its form (Sun → ATP)
 - Life create local order (but increases the disorder ness elsewhere in the universe, REMEMBER “lost heat”)
2. Driving force is in the direction of lower free energy ($\Delta G = -ve$).
3. Driving force does not mean that the process will happen fast. It depends on the path connecting the reactant and product (Activation barrier).
4. Enzymes lowers the activation barrier and speed-up the process to attain equilibrium.
5. Catabolic processed produce energy (down-hill). Anabolic processes require energy (Up-hill).
6. Metabolic pathway is a linked series of chemical reactions occurring within a cell involving enzymes.



Next: Summary from key concepts & Nutrients